

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4-AT1 - Scenario-Based Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4-AT1- Scenario-Based Questions	Assessment:	Assessment Tool 1– Scenario-Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze real-world business data using confidence intervals, hypothesis testing, test statistics, and p-value approaches to evaluate claims and support data-driven industry decisions.		
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Section / Batch:	AI-DS	Date:	16/08/26

Code of Conduct

I B Tharun Kumar certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *B Tharun Kumar*

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	

Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Statistics Assessment — Analytical Solutions with Formula Derivation

Confidence Intervals & Hypothesis Testing (Problems 6–10)

Problem 1 — Online Learning Platform

An online learning platform measures the completion time of a course for 100 students. The mean completion time is 14 days and the standard deviation is 3 days. Construct a 95% confidence interval for the true mean completion time.

Step 1 — Identify the Population and Sample

Population = all students who have ever taken (or will take) this course on the platform.

Sample = the 100 students whose completion times were actually measured.

Step 2 — Identify the Parameter of Interest

μ = the true (unknown) population mean course completion time, measured in days. We are trying to estimate this using the sample data.

Step 3 — State the Point Estimate

The sample mean $\bar{x} = 14$ days is our single best guess (point estimate) of μ . However, a single number gives no sense of how reliable that guess is — this is why we build a confidence interval, which gives a plausible range for μ along with a stated confidence level.

Step 4 — Write and Understand the Formula

$$CI = \bar{x} \pm z(\alpha/2) \times (s / \sqrt{n})$$

- $\bar{x}$ → the sample mean (point estimate of the unknown population mean μ).
- $z(\alpha/2)$ → the critical value from the standard normal (Z) distribution corresponding to the chosen confidence level. It tells us how many standard errors to move away from $\bar{x}$ so that, under repeated sampling, the interval captures the true mean the stated percentage of the time. For 95% confidence, $\alpha = 1 - 0.95 = 0.05$, so $\alpha/2 = 0.025$, and $z(0.025) = 1.96$ (from the standard normal table).
- s → the sample standard deviation, measuring how spread out individual completion times are around the sample mean.
- n → the sample size (number of students measured). We use n because it is large ($n = 100 \geq 30$), so by the Central Limit Theorem the sampling distribution of $\bar{x}$ is approximately normal, allowing us to use z instead of t .
- $s/\sqrt{n}$ → this is the Standard Error (SE) of the mean. It measures how much the sample mean $\bar{x}$ is expected to vary from sample to sample. Notice that as n increases, SE decreases — larger samples give more precise (narrower) estimates of μ .
- $z(\alpha/2) \times SE$ → this product is called the Margin of Error (E). It is the amount added and subtracted from $\bar{x}$ to build the lower and upper bounds of the interval.

Step 5 — Substitute Values and Solve (Analytically)

Given: $n = 100$, $\bar{x} = 14$ days, $s = 3$ days, confidence level = 95% $\Rightarrow \alpha = 0.05$, $z(0.025) = 1.96$.

Step A — Compute the Standard Error: $SE = s / \sqrt{n} = 3 / \sqrt{100} = 3 / 10 = 0.3$.

Step B — Compute the Margin of Error: $E = z(\alpha/2) \times SE = 1.96 \times 0.3 = 0.588$.

Step C — Compute the interval bounds: Lower bound = $\bar{x} - E = 14 - 0.588 = 13.412$;
Upper bound = $\bar{x} + E = 14 + 0.588 = 14.588$.

Step 6 — Final Confidence Interval

95% CI = (13.412 days, 14.588 days)

Step 7 — Interpretation / Conclusion

We are 95% confident that the true mean completion time for the entire student population lies between 13.412 and 14.588 days. In the frequentist sense, this means: if we repeated this sampling process ($n = 100$ students) many times and built a CI each time, about 95% of those intervals would contain the true population mean. This range helps the platform set realistic completion-time expectations and identify whether course pacing needs adjustment.

Problem 2 — Hospital Patient Satisfaction

A hospital surveys 400 randomly selected patients, and 340 report that they are satisfied with the service. Construct a 95% confidence interval for the proportion of satisfied patients.

Step 1 — Identify the Population and Sample

Population = all patients treated by the hospital. Sample = the 400 randomly selected patients who were surveyed.

Step 2 — Identify the Parameter of Interest

p = the true (unknown) population proportion of all patients who are satisfied with the hospital's service.

Step 3 — State the Point Estimate

The sample proportion $\hat{p} = x/n = 340/400 = 0.85$ (85%) is our point estimate of the true proportion p . As with the mean, this single value needs a margin of uncertainty attached, hence the confidence interval.

Step 4 — Write and Understand the Formula

$$CI = \hat{p} \pm z(\alpha/2) \times \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

- $\hat{p} \rightarrow$ the sample proportion (number of successes x divided by sample size n): $\hat{p} = x/n$.
- $z(\alpha/2) \rightarrow$ same role as before — the critical Z-value for the chosen confidence level. For 95% confidence, $z(0.025) = 1.96$.
- $\hat{p}(1 - \hat{p}) \rightarrow$ this term comes from the variance of a binomial proportion; it is largest when $\hat{p} = 0.5$ and smaller as $\hat{p}$ moves toward 0 or 1, reflecting less variability near the extremes.
- $\sqrt{[\hat{p}(1 - \hat{p})/n]} \rightarrow$ this is the Standard Error of the proportion (SE). It measures how much $\hat{p}$ is expected to fluctuate from sample to sample of size n .
- $z(\alpha/2) \times SE \rightarrow$ the Margin of Error (E), added and subtracted from $\hat{p}$ to form the interval.

Step 5 — Substitute Values and Solve (Analytically)

Given: $n = 400$, $x = 340$, so $\hat{p} = 340/400 = 0.85$; confidence level = 95% $\Rightarrow z(0.025) = 1.96$.

Step A — Compute the Standard Error: $SE = \sqrt{[\hat{p}(1 - \hat{p})/n]} = \sqrt{[(0.85 \times 0.15)/400]} = \sqrt{(0.1275/400)} = \sqrt{0.00031875} \approx 0.01785$.

Step B — Compute the Margin of Error: $E = z(\alpha/2) \times SE = 1.96 \times 0.01785 \approx 0.0350$.

Step C — Compute the interval bounds: Lower bound = $0.85 - 0.0350 = 0.8150$; Upper bound = $0.85 + 0.0350 = 0.8850$.

Step 6 — Final Confidence Interval

95% CI = (0.8150, 0.8850), i.e. (81.50%, 88.50%)

Step 7 — Interpretation / Conclusion

We are 95% confident that the true proportion of all patients who are satisfied with the hospital's service lies between 81.5% and 88.5%. This interval allows hospital management to gauge overall service quality and track whether satisfaction genuinely differs from a target benchmark (e.g., 90%) in future audits.

Problem 3 — Telecom Monthly Data Usage

A telecom company wants to estimate the average monthly data usage of its customers. A sample of 225 customers has a mean usage of 16 GB and a standard deviation of 5 GB. Construct a 99% confidence interval and interpret the result.

Step 1 — Identify the Population and Sample

Population = all customers of the telecom company. Sample = the 225 customers whose monthly data usage was measured.

Step 2 — Identify the Parameter of Interest

μ = the true (unknown) population mean monthly data usage, measured in GB.

Step 3 — State the Point Estimate

The sample mean $\bar{x} = 16$ GB is the point estimate of μ , the average monthly usage across all customers.

Step 4 — Write and Understand the Formula

$$CI = \bar{x} \pm z(\alpha/2) \times (s / \sqrt{n})$$

- $\bar{x}$ → sample mean (point estimate of μ) = 16 GB.
- $z(\alpha/2)$ → Z critical value for the chosen confidence level. This time the confidence level is 99%, so $\alpha = 1 - 0.99 = 0.01$, and $\alpha/2 = 0.005$. Looking this up in the standard

normal table gives $z(0.005) = 2.576$. Notice this is larger than the 1.96 used for 95% confidence — a higher confidence level requires a wider interval, hence a larger multiplier.

- $s \rightarrow$ sample standard deviation = 5 GB, describing spread of individual usage values.
- $n \rightarrow$ sample size = 225, which is large enough (≥ 30) to justify using the z -distribution.
- $s/\sqrt{n} \rightarrow$ Standard Error (SE), the expected variability of the sample mean $\bar{x}$ across repeated samples.
- $z(\alpha/2) \times SE \rightarrow$ Margin of Error (E), the buffer added/subtracted around $\bar{x}$.

Step 5 — Substitute Values and Solve (Analytically)

Given: $n = 225$, $\bar{x} = 16$ GB, $s = 5$ GB, confidence level = 99% $\Rightarrow z(0.005) = 2.576$.

Step A — Compute the Standard Error: $SE = s/\sqrt{n} = 5/\sqrt{225} = 5/15 = 0.3333$.

Step B — Compute the Margin of Error: $E = z(\alpha/2) \times SE = 2.576 \times 0.3333 \approx 0.8587$.

Step C — Compute the interval bounds: Lower bound = $16 - 0.8587 = 15.1413$; Upper bound = $16 + 0.8587 = 16.8587$.

Step 6 — Final Confidence Interval

99% CI = (15.1413 GB, 16.8587 GB)

Step 7 — Interpretation / Conclusion

We are 99% confident that the true average monthly data usage across all customers lies between 15.14 GB and 16.86 GB. Because the confidence level is higher than a typical 95%, this interval is intentionally wider — reflecting the added certainty required. The telecom company can use this range for network capacity planning and to design appropriately sized data plans that match typical usage.

Problem 4 — Customer-Support Resolution Time

A company claims that the average time taken by its customer-support team to resolve a complaint is 24 hours. A sample produces a 95% confidence interval of (25.2, 27.8) hours. Based only on this confidence interval, would you reject the company's claim? Explain.

Step 1 — Identify the Population and Sample

Population = all customer complaints handled by the support team. Sample = the complaints included in the study that produced the given confidence interval.

Step 2 — Identify the Parameter of Interest

μ = the true (unknown) population mean resolution time, in hours. The company claims $\mu = 24$ hours.

Step 3 — State the Hypotheses

Null Hypothesis (H_0): $\mu = 24$ hours (the company's claim is correct — average resolution time really is 24 hours).

Alternative Hypothesis (H_1): $\mu \neq 24$ hours (the true average resolution time differs from the claimed 24 hours) — this is a two-tailed test.

Step 4 — Understand the Decision Rule (Formula/Logic)

**If μ_0 (the claimed/hypothesized value) lies INSIDE the confidence interval
→ Fail to Reject H_0**

If μ_0 lies OUTSIDE the confidence interval → Reject H_0

- Why this works: a $(1 - \alpha) \times 100\%$ confidence interval contains precisely the set of hypothesized values μ_0 that a two-tailed test at significance level α would NOT reject.

In other words, the CI is the 'acceptance region' for H_0 expressed directly in the units of μ , rather than in test-statistic units.

- Equivalently, any μ_0 value that falls outside the CI would, if tested formally, produce a test statistic large enough that its two-tailed p-value would be less than α — leading to rejection of H_0 .
- This lets us test a claim quickly using only a previously-constructed CI, without recomputing a t- or z-statistic from raw data.

Step 5 — Apply the Rule to the Given Interval

Given: claimed value $\mu_0 = 24$ hours; 95% CI = (25.2 hours, 27.8 hours); significance level $\alpha = 1 - 0.95 = 0.05$.

Step A — Check whether $\mu_0 = 24$ lies within the interval (25.2, 27.8).

Step B — Since $24 < 25.2$ (the lower bound of the interval), the value 24 does NOT lie inside the interval — it falls below the entire range.

Step 6 — Decision

Because the claimed value (24 hours) lies outside the 95% confidence interval (25.2, 27.8), we REJECT H_0 at the 5% significance level.

Step 7 — Industry-Oriented Conclusion

There is statistically significant evidence, at the 5% level, that the true average complaint-resolution time is NOT 24 hours. Moreover, since the entire interval (25.2 to 27.8 hours) lies above 24, the evidence specifically suggests the actual average resolution time is longer than claimed. The company should revisit its stated service-level commitment or investigate ways to speed up complaint resolution to match its claim.

Problem 5 — Manufacturing — Packaged Product Weight

A manufacturer claims that the average weight of a packaged product is 500 grams. A 95% confidence interval based on a sample is (496, 502) grams. Is the claimed value of 500 grams consistent with the sample evidence? Explain your answer.

Step 1 — Identify the Population and Sample

Population = all packaged products produced by the manufacturer. Sample = the units included in the study that produced the given confidence interval.

Step 2 — Identify the Parameter of Interest

μ = the true (unknown) population mean package weight, in grams. The manufacturer claims $\mu = 500$ g.

Step 3 — State the Hypotheses

Null Hypothesis (H_0): $\mu = 500$ g (the manufacturer's claim is correct).

Alternative Hypothesis (H_1): $\mu \neq 500$ g (the true average weight differs from the claimed 500 g) — two-tailed test.

Step 4 — Understand the Decision Rule (Formula/Logic)

**If μ_0 (the claimed/hypothesized value) lies INSIDE the confidence interval
→ Fail to Reject H_0**

If μ_0 lies OUTSIDE the confidence interval → Reject H_0

- As explained for Problem 9, the confidence interval represents the full set of values that would NOT be rejected by a two-tailed hypothesis test at significance level $\alpha = 1 - \text{confidence level}$.

- So checking whether μ_0 falls inside or outside the interval is mathematically equivalent to carrying out the full hypothesis test — it simply reads the decision directly off the interval.

Step 5 — Apply the Rule to the Given Interval

Given: claimed value $\mu_0 = 500$ g; 95% CI = (496 g, 502 g); significance level $\alpha = 1 - 0.95 = 0.05$.

Step A — Check whether $\mu_0 = 500$ lies within the interval (496, 502).

Step B — Since $496 < 500 < 502$, the value 500 DOES lie inside the interval.

Step 6 — Decision

Because the claimed value (500 g) lies inside the 95% confidence interval (496, 502), we FAIL TO REJECT H_0 at the 5% significance level.

Step 7 — Industry-Oriented Conclusion

There is not enough statistical evidence, at the 5% level, to conclude that the true average package weight differs from the claimed 500 grams. The sample data are consistent with the manufacturer's claim, so quality-control processes appear to be functioning as expected — no corrective action is indicated based on this evidence alone.